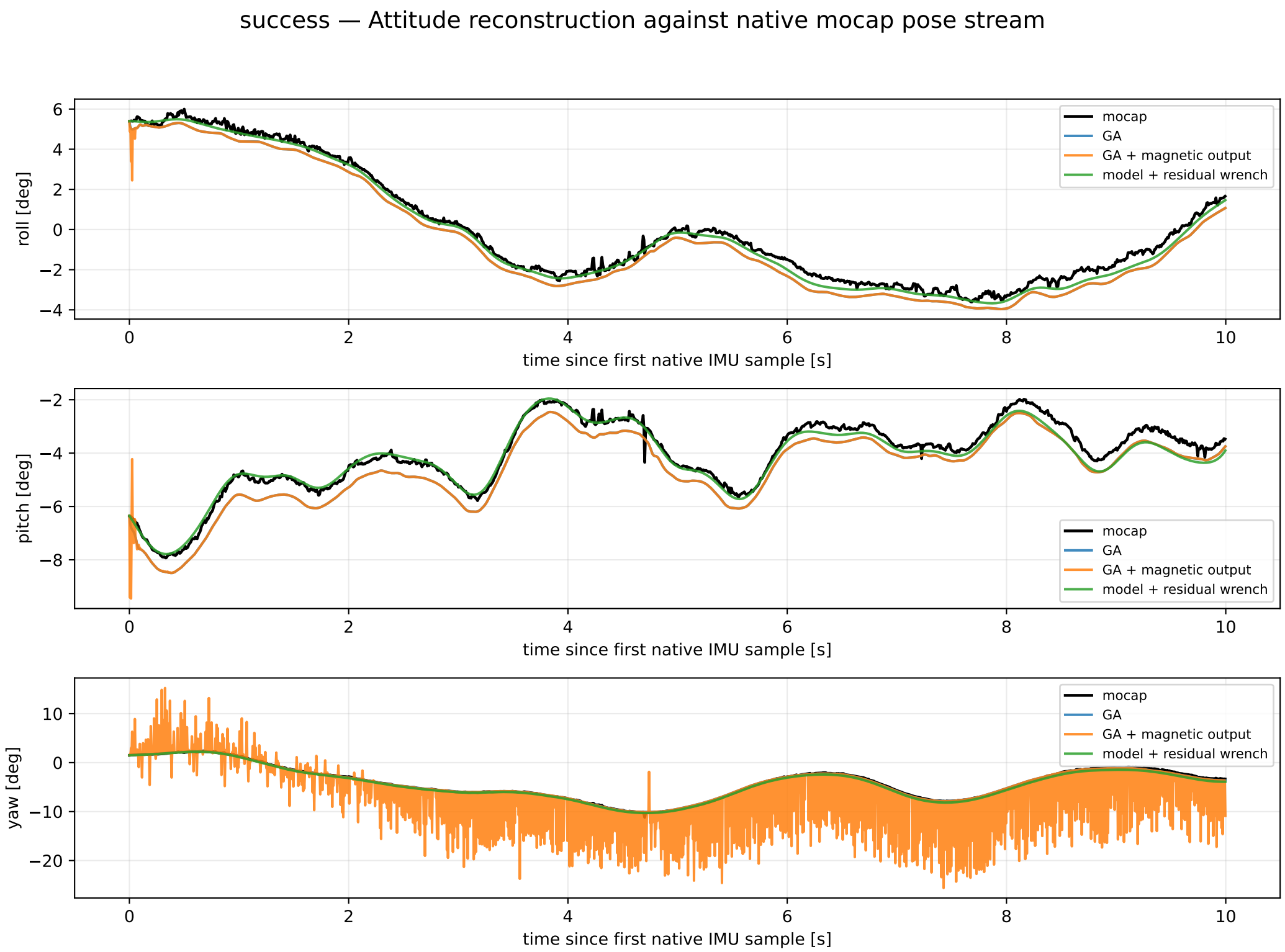
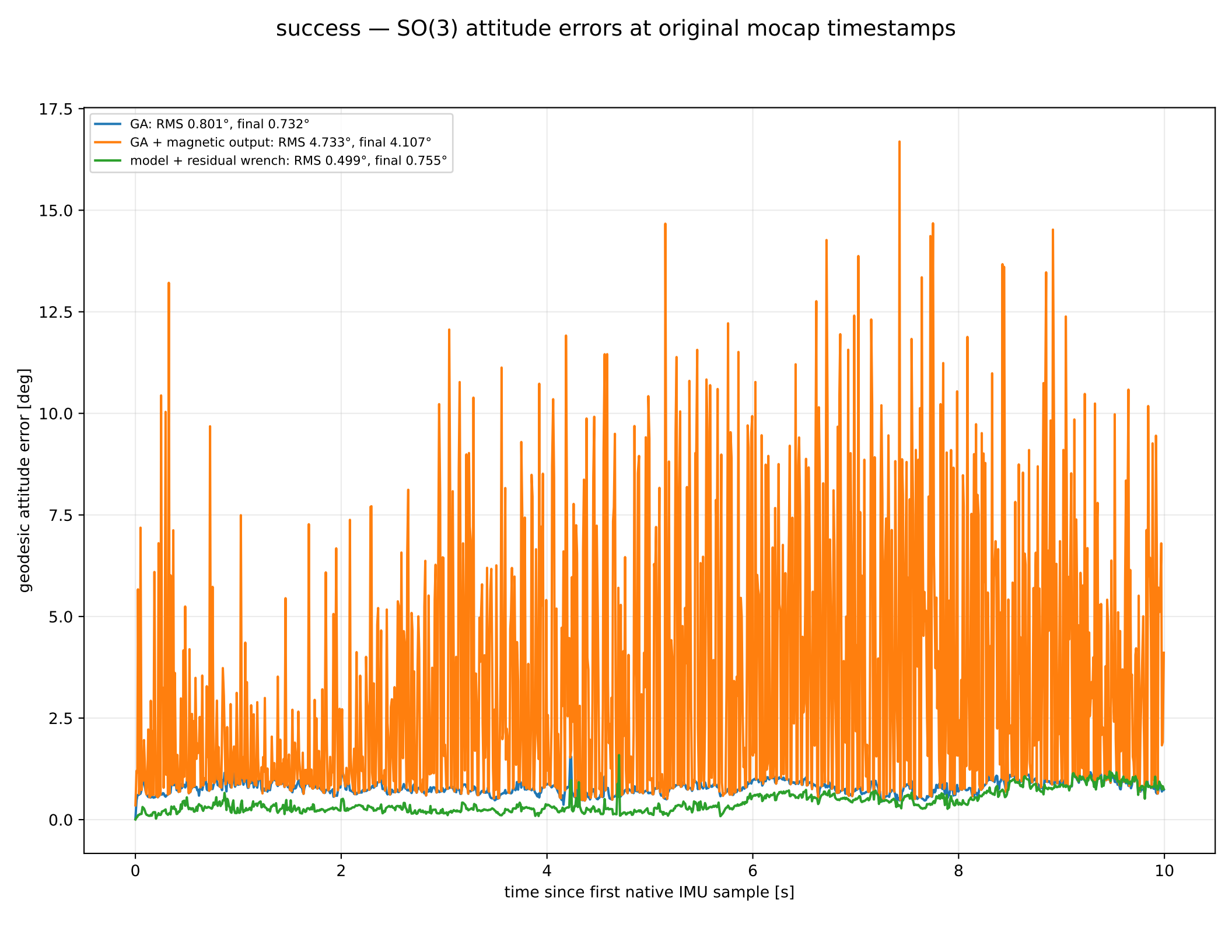
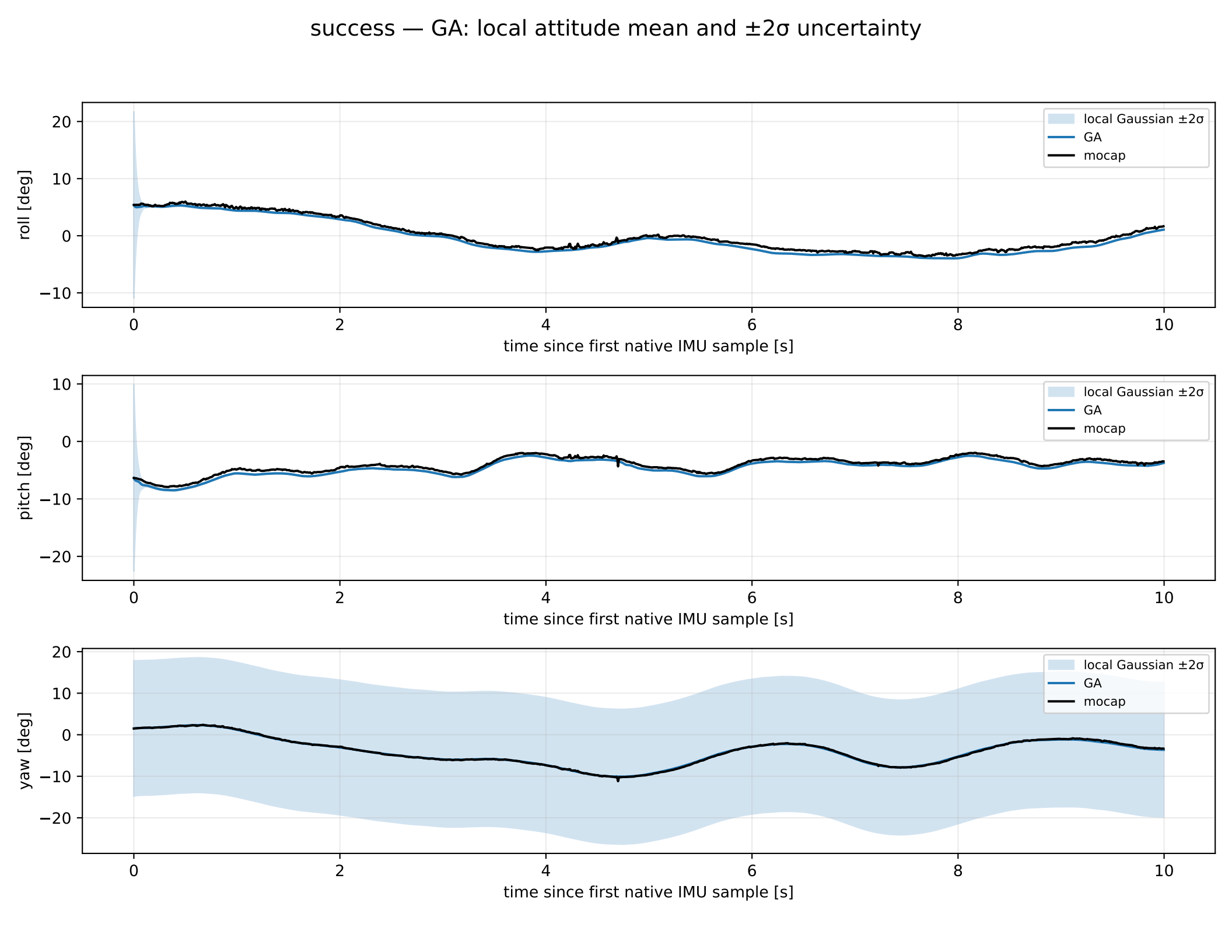
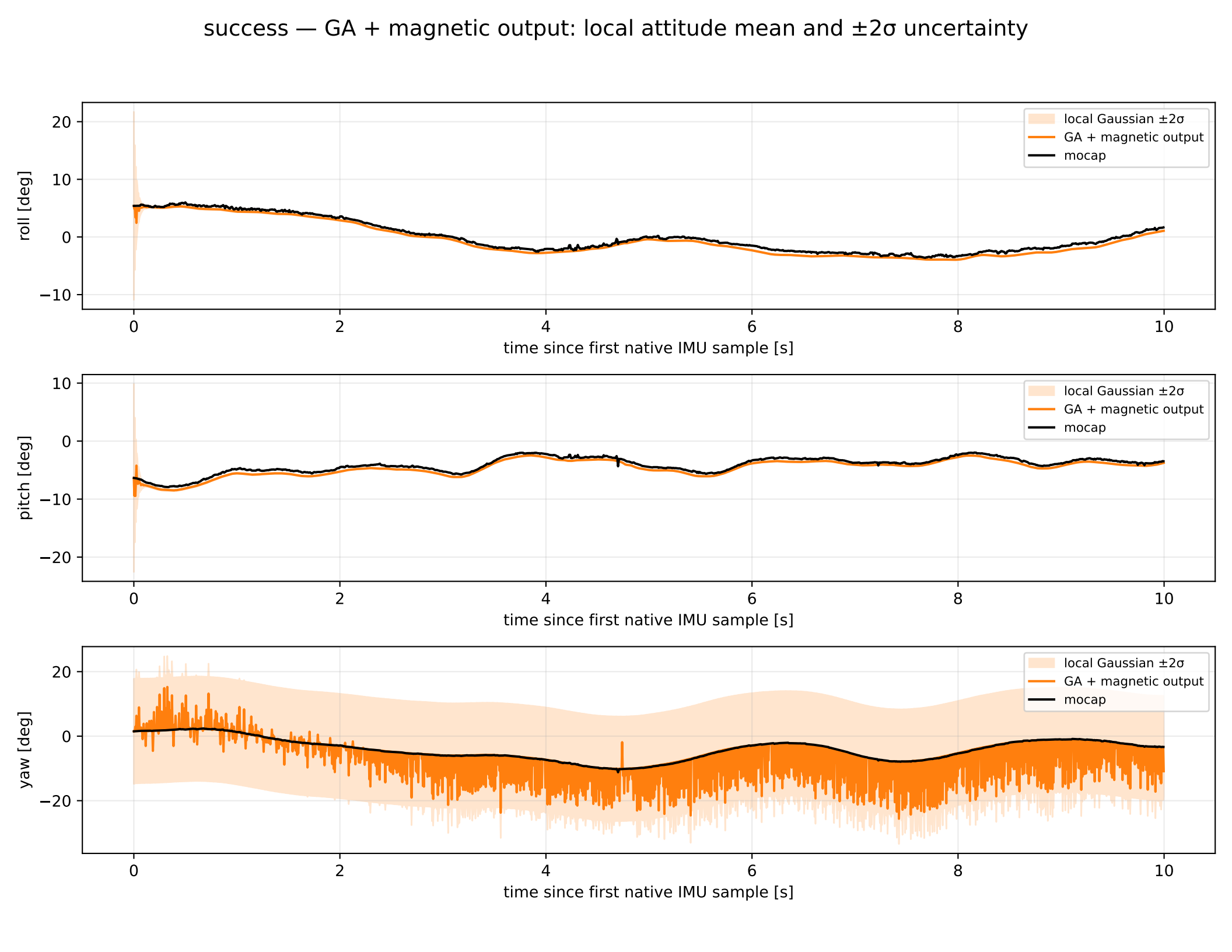
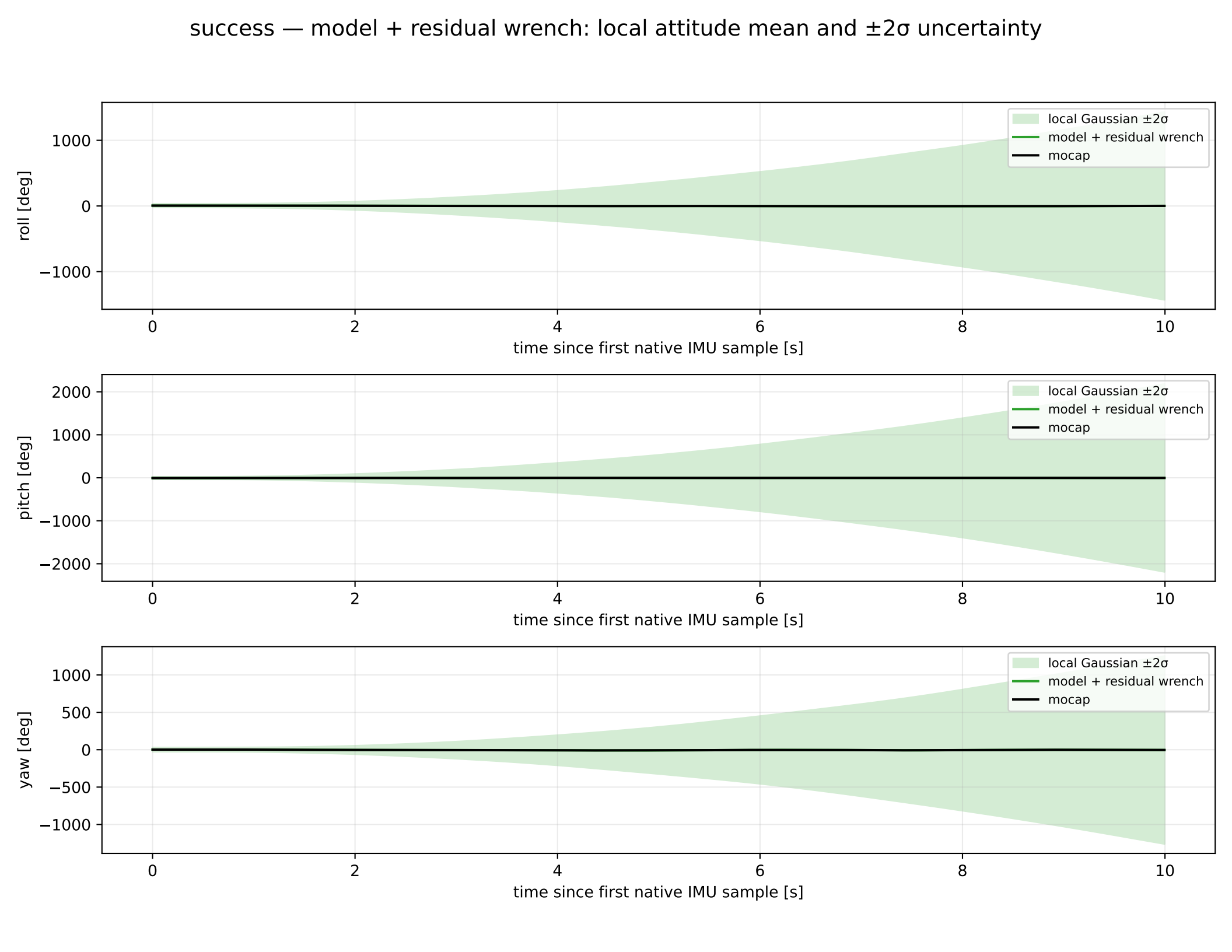

success — Attitude reconstruction against native mocap pose stream
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
mocap
GA
GA + magnetic output
model + residual wrench

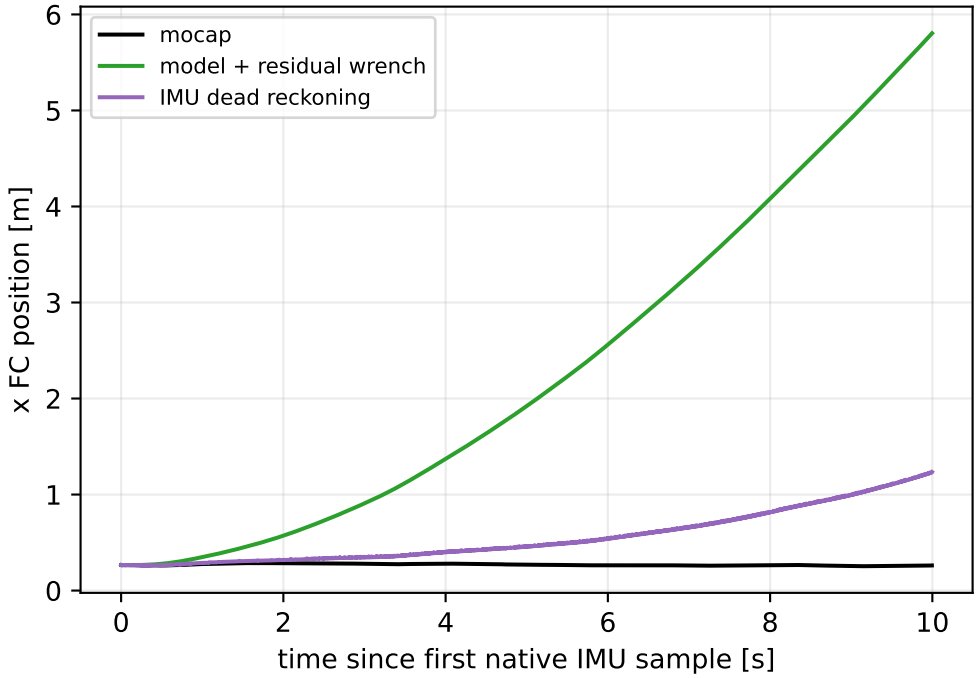

success — SO(3) attitude errors at original mocap timestamps
GA: RMS 0.801°, final 0.732°
GA + magnetic output: RMS 4.733°, final 4.107°
model + residual wrench: RMS 0.499°, final 0.755°
geodesic attitude error [deg]
time since first native IMU sample [s]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
2
4
6
8
10

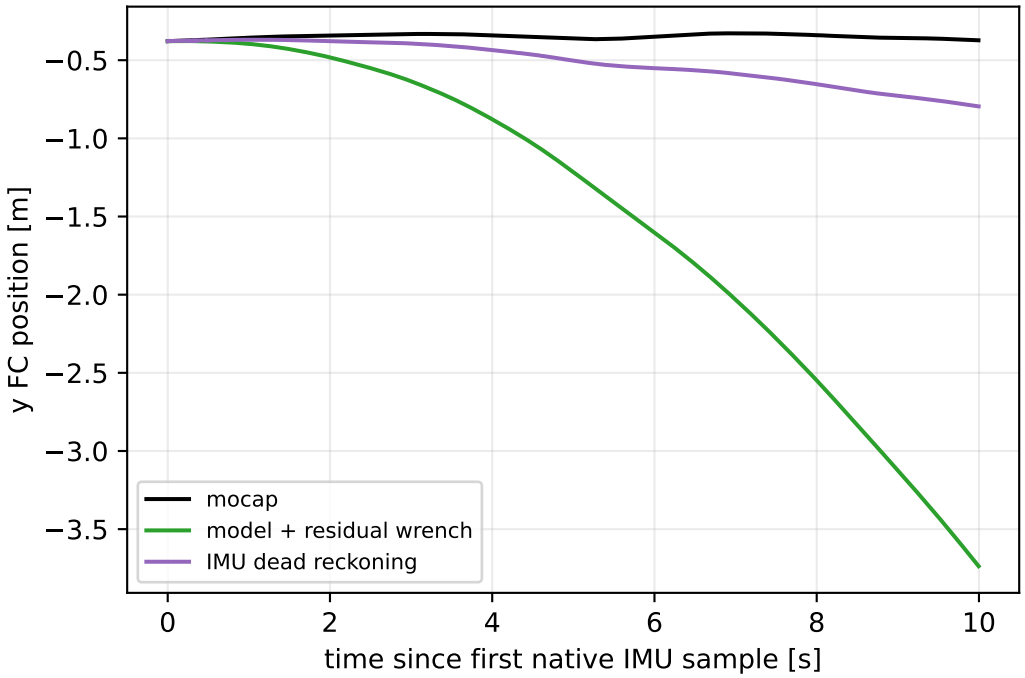

success — GA: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA
mocap

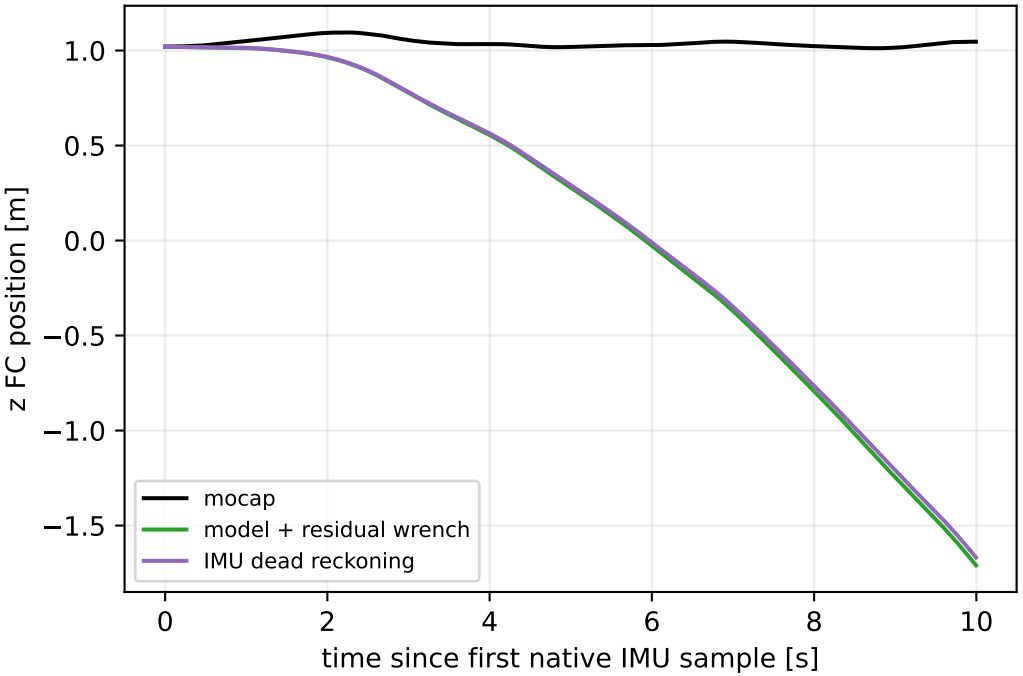

success — GA + magnetic output: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA + magnetic output
mocap

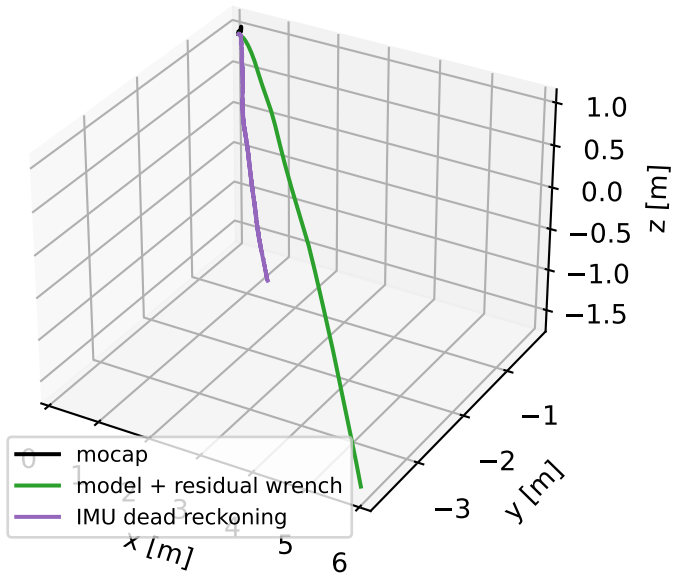

success — model + residual wrench: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
model + residual wrench
mocap

success — Position reconstruction (mean trajectories)



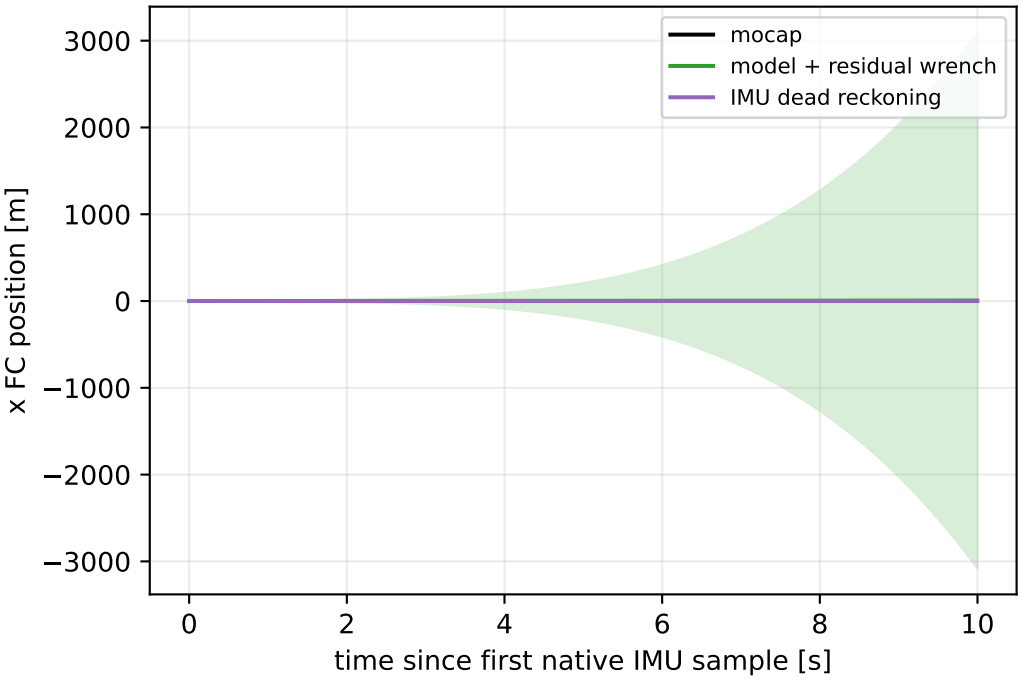


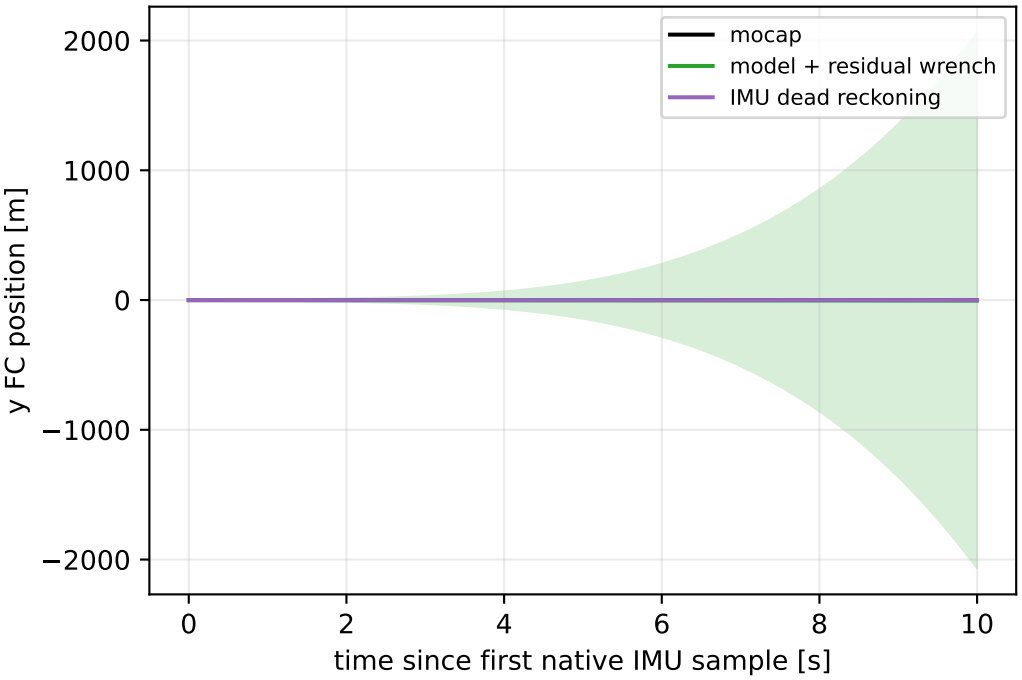


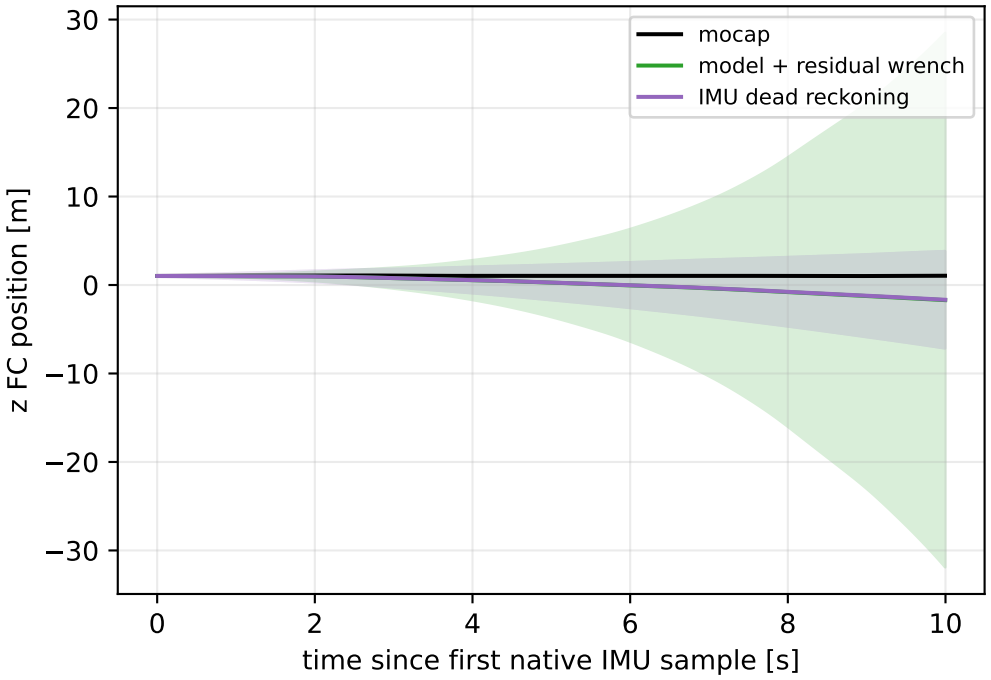


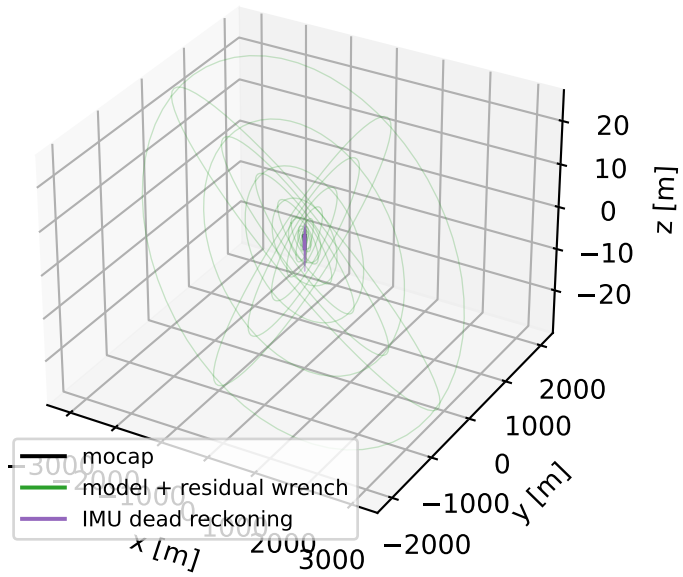
Residual-wrench model replay and IMU integration are not reset to later mocap.

success — Position reconstruction with full propagated ±2σ spread









FC origin; local Gaussian linearization. Ellipsoids: deterministic 2σ factor-SVD wireframes, not samples.